

Q4 Do we need IV estm?

(2b)

Nu-Hausman test of exogeneity

H_0 : Regressors are asymptotically uncorrelated with ~~error~~ ^{error}. ag. H_A : H_0 not true

$$\Leftrightarrow H_0: \lim_{T \rightarrow \infty} \frac{X'E}{T} = 0 \text{ (and } \lim_{T \rightarrow \infty} \frac{X'X}{T} = Q') \text{ ag. } H_A: \lim_{T \rightarrow \infty} \frac{X'E}{T} \neq 0$$

Model: $Y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \beta_3 x_{t3} + \epsilon_t$, suppose only x_3 endogeneity

Then $x^{(1)} = (1, x_1, x_2)$, $x^{(2)} = (x_3)$, $Z = (1, x_1, x_2, z_1, z_2)$ over identified case.

Step 1 to OLS estm of the instrumenting regressor/s by other regressors & instruments

$$\text{ie, } x_{t3} = \alpha_0 + \alpha_1 x_{t1} + \alpha_2 x_{t2} + \gamma_1 z_{t1} + \gamma_2 z_{t2} + u_t, \text{ obtain } \hat{\alpha}'s, \hat{\gamma}'s \text{ \& } \hat{u}_t's$$

Step 2 use a 'variable addition test' to test significance of $\hat{u}_t$.

$$\text{ie, } y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \beta_3 x_{t3} + \theta \hat{u}_t + v_t \quad (v_t \text{ is free of } x_3's \text{ effect as } \hat{u}_t \text{ has been added})$$

Step 3 $H_0: \theta = 0$ ag $H_A: \theta \neq 0$.

Usual t-test rejects H_0

$\Rightarrow x_3$ asymptotically correlated with ϵ
use $\hat{\beta}_{IV}$.

Remarks i) Have to guess which var/s could be correlated with ϵ
(Nu-Hausman test is valid iff other reg./instruments are asymptotically uncorrelated with ϵ)

ii) What if $\exists$ more than 1 endogeneity var.

Soln Choose at least as many instruments

Δ Repeat step 1 for every endogeneity var.

Include residuals from every such model & test

$H_0: \theta_1 = \dots = \theta_k$ by F-test.

Final qz Are the chosen instruments valid?

Sargan's test (can be applied only in over identified case)

$$H_0: \lim_{T \rightarrow \infty} \frac{Z'E}{T} = 0 \text{ ag } H_A: \lim_{T \rightarrow \infty} \frac{Z'E}{T} \neq 0$$

(27)

$$y_{Tx1} = X_{kx1}^{Txk} \beta + \epsilon_{Tx1}, \quad \begin{matrix} X_{kx1}^{Txk} : \text{regressors} \\ Z_{Txp} : \text{instruments, } p > k \end{matrix}$$

$$\hat{\beta}_{IV} = (X' P_Z X)^{-1} X' P_Z y, \quad P_Z = Z(Z'Z)^{-1}Z'$$

$$\hat{\epsilon} = y - \hat{y} = y - X\hat{\beta}_{IV}$$

If $\text{Cov}(\epsilon) = \sigma^2 I$, $\text{Cov}(\hat{\beta}_{IV}) = \sigma^2 (X' P_Z X)^{-1}$, $\hat{\text{Cov}}(\hat{\beta}_{IV}) = \hat{\sigma}^2 (X' P_Z X)^{-1}$,
 Hence $\hat{\sigma}_{\text{unb}}^2 = \frac{\hat{\epsilon}' \hat{\epsilon}}{T-k}$

(Generalized est., $\hat{\beta}_{IV}$ obtained by minimizing $Q(\beta) = (y - X\beta)' P_Z (y - X\beta)$ wrt β).

Test stat. $\chi_{SM}^2 = \frac{Q(\hat{\beta}_{IV})}{\hat{\sigma}^2} = \frac{\hat{\epsilon}' P_Z \hat{\epsilon}}{\hat{\sigma}^2} \sim \chi_{p-k}^2$ under H_0 .
 (Sargan's test of misspecification).

$$W: \text{Obd} \chi_{SM}^2 > \chi_{p-k; 1-\alpha}^2$$

Underlying Distributions?

$$y_t = x_t' \beta + \epsilon_t, \quad t=1(1)T$$

with ϵ_t iid $N(0, \sigma^2)$.

Assume $E(x_t \epsilon_t) = 0 \quad \forall t$,

ie, weak exogeneity of regressors.

Regularity condns (for ϵ_t regressors):

$$(*) \quad \begin{cases} \text{plim}_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T x_t x_t' = Q, \text{ p.d.} \\ \text{plim}_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T x_t \epsilon_t = 0 \end{cases}$$

IV estn required, when $\text{plim}_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T x_t \epsilon_t \neq 0$

Regularity condns (for instruments)

$$(**) \quad \begin{cases} \text{plim}_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T z_t x_t' = Q_{zx}, \text{ n.s.} \\ \text{plim}_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T z_t z_t' = Q_{zz}, \text{ p.d.} \end{cases}$$

Alternative to matrix representation -

$$y = X\beta + \epsilon,$$

$$\epsilon \sim N_T(0, \sigma^2 I_T)$$

and so on...

Then, (follows from the Haur-Wald theo.) / let z_t be a vector of r.v.s s.t. ϵ_t be a seq. of iid r.v.s with zero mean, finite var σ^2 & (2) hold, then (18)

i) $\frac{1}{\sqrt{T}} \sum_{t=1}^T z_t \epsilon_t \xrightarrow{d} N(0, \sigma^2 \Sigma_z)$

ii) $\frac{1}{\sqrt{T}} \sum_{t=1}^T z_t^2 \epsilon_t \xrightarrow{d} N(0, \sigma^2 \Sigma_{zz})$

so that $\hat{\beta}_N = \beta + (z'X)^{-1} z' \epsilon$

then $(\hat{\beta}_N - \beta) \sqrt{T} = \left(\frac{z'X}{T} \right)^{-1} \left(\frac{z'\epsilon}{\sqrt{T}} \right)$

$\Rightarrow \sqrt{T}(\hat{\beta}_N - \beta) \xrightarrow{d} N(0, \sigma^2 \Sigma_z^{-1} \Sigma_{zz} \Sigma_z^{-1})$ using (*) & $\epsilon \sim N(0, \sigma^2 I_T)$

IV Estn in system of Eqs

$y_i = x_i \beta_i + \epsilon_i, i=1(1)N, y_i \text{ is } T \times 1, x_i \text{ is } T \times k_i$

Combining, $y = X\beta + \epsilon$
 $TN \times 1 \quad TN \times 1 \quad TN \times K \quad TN \times 1$
 $y = \begin{pmatrix} y_1 \\ \vdots \\ y_i \\ \vdots \\ y_N \end{pmatrix}, X = \begin{pmatrix} x_1 & 0 & \dots & 0 \\ \vdots & & & \\ 0 & \dots & 0 & x_N \end{pmatrix}, \beta = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_N \end{pmatrix}$

$E(\epsilon|x) = 0, E(\epsilon\epsilon'|x) = \Omega \otimes I_T = V$, say, $\Omega = ((\omega_{ij})) = E(\epsilon_i \epsilon_i')$
 $N \times 1$

If $Cov(x, \epsilon) \neq 0$ even asymptotically, then $\hat{\beta}_{OLS}$ (or $\hat{\beta}_{OLS}$) inconsistent.

$\hat{\beta}_{IV} = (z'X)^{-1} z'y$ (corr. to $\hat{\beta}_{OLS}$)

or $\hat{\beta}_{GIVE} = (z'(\Omega' \otimes I_T))^{-1} z'(\Omega' \otimes I_T)y$ (corr. to $\hat{\beta}_{OLS}$)
 (Generalised IV estimator)

are consistent

Corr. to $\hat{\beta}_{2SLS} = (\hat{X}'\hat{X})^{-1} \hat{X}'y$ (Generalised 2SLS est.)

is $\hat{\beta}_{3SLS} = (\hat{X}'(\Omega' \otimes I)\hat{X})^{-1} [\hat{X}'(\Omega' \otimes I)y]$

where $\hat{X} = \begin{pmatrix} \hat{x}_1 & 0 & \dots & 0 \\ 0 & \hat{x}_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \hat{x}_N \end{pmatrix}$

eqn by eqn OLS estn. h
 get $\hat{x}_i$

$x_i = z_i \pi_i + u_i \quad \forall i=1(1)N$

so that $\hat{x}_i = z_i \hat{\pi}_i$